

Tutorial: JUnit

Suggested answers

Q1. Suppose you have the following class which you want to test:

```
public class Calculation {  
    public static int findMax(int arr[]) {  
        int max=arr[0];  
        for(int i=1;i<arr.length;i++){  
            if(max<arr[i])  
                max=arr[i];  
        }  
        return max;  
    }  
}
```

Here findMax() is a function that accepts an array of integers as its input parameter. It returns the largest value among the values in the input array. Write a Junit unit test for the findMax() function.

The requirement is: Use an array with values 1, 3, 4, 2 as the test case. Here the data type of 1 is int, of 3 is int, of 4 is int, and of 2 is int.

Example answer:

```
public class TestFindMax {  
  
    @Test    // Important  
  
    public void testFindMax(){  
        assertEquals(4, Calculation.findMax(new int[]{1,3,4,2}));  
    }  
}
```

Q2. Suppose you have the following Calculator class which you want to test.

```
public class Calculator {  
    public int multiply(int a, int b) {  
        return a * b;  
    }  
}
```

Fill up the following test class to complete the testing for the above Calculator class.

```

class CalculatorTest {
    ...
    void setUp() {
        ...
    }
    void testMultiply() {
        ...
    }
}

```

Additional requirements are:

- Use $4 * 5 = 20$ as the test case
- The test should run 5 times
- to import only the junit packages that are needed for this question.

Example answers:

1. To import junit packages for using :

assertEquals()

@RepeatedTest

@DisplayName

```
import org.junit.jupiter.api.DisplayName;
```

```
import org.junit.jupiter.api.RepeatedTest;
```

```
import static org.junit.jupiter.api.Assertions.assertEquals; //
```

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(<https://junit.org/junit5/docs/current/user-guide/> Section 2. Writing tests, Section 2.4 display names, and Section 2.16 repeated tests)

2. To complete the class

```
class CalculatorTest {  
  
    Calculator calculator = new Calculator();  
  
    // The requirement says  
    // to run 5 times  
    // which annotation to choose?  
  
    @RepeatedTest(5)           // important  
    @DisplayName("Simple multiplication should work") // example text here  
    void testMultiply() {  
        assertEquals(20, calculator.multiply(4, 5),  
            "Regular multiplication should work");  
    }  
    // important: 1. use the right assertEquals() function ; 2. function parameters correct  
}
```